

# PHYSICS 531 Honor Project

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August 18, 2025

## 1 Eigenvalues of Dirac-like Hamiltonians in 1-3D Space

To determine the energy eigenvalues of the Hamiltonians specified in Problem 1, we transition from a position-space representation to a momentum-space representation using the Fourier transform. The wavefunction  $\Psi(\mathbf{r})$  is expressed as a superposition of plane waves:

$$\Psi(\mathbf{r}) = \int \frac{d^D p}{(2\pi)^D} e^{i\mathbf{p}\cdot\mathbf{r}} \tilde{\Psi}(\mathbf{p}), \quad (1)$$

where  $D$  is the spatial dimension (1, 2, or 3),  $\mathbf{p}$  is the momentum vector, and  $\tilde{\Psi}(\mathbf{p})$  is the momentum-space wavefunction. In this basis, the momentum operator  $\hat{\mathbf{p}} = -i\nabla$  (with  $\hbar = 1$  as per the problem's convention) acting on  $\Psi(\mathbf{r})$  becomes multiplication by the vector  $\mathbf{p}$  for  $\tilde{\Psi}(\mathbf{p})$ .

The eigenvalue equation for the Hamiltonian  $\hat{H}$  in position space,  $\hat{H}\Psi(\mathbf{r}) = E\Psi(\mathbf{r})$ , transforms into an algebraic eigenvalue problem in momentum space:

$$H(\mathbf{p})\tilde{\Psi}(\mathbf{p}) = E(\mathbf{p})\tilde{\Psi}(\mathbf{p}). \quad (2)$$

Here,  $H(\mathbf{p})$  is the Hamiltonian matrix expressed in terms of momentum components  $\mathbf{p}$ , and  $E(\mathbf{p})$  are the corresponding energy eigenvalues. These eigenvalues are found by diagonalizing the matrix  $H(\mathbf{p})$ .

For the Dirac-like Hamiltonians considered, which are linear in momentum and include a mass term, the energy eigenvalues  $E(\mathbf{p})$  typically follow a relativistic dispersion relation. Assuming the characteristic velocity  $v = 1$  (as implied by "speed of light is set to unity" in the problem description), the general form of these eigenvalues is:

$$E_{\pm}(\mathbf{p}) = \pm\sqrt{|\mathbf{p}|^2 + m^2}, \quad (3)$$

where  $m$  is the mass parameter, and  $|\mathbf{p}|$  is the magnitude of the momentum vector. The  $\pm$  signs distinguish between particle and anti-particle solutions (or upper and lower energy bands).

We can specify this for each dimension:

**1D Case:** In one spatial dimension, the momentum is  $\mathbf{p} = (p_x)$ , so  $|\mathbf{p}|^2 = p_x^2$ . The energy eigenvalues are:

$$E_{\pm}(p_x) = \pm\sqrt{p_x^2 + m^2}. \quad (4)$$

**2D Case:** In two spatial dimensions, the momentum is  $\mathbf{p} = (p_x, p_y)$ , so  $|\mathbf{p}|^2 = p_x^2 + p_y^2$ . The energy eigenvalues are:

$$E_{\pm}(\mathbf{p}) = \pm \sqrt{p_x^2 + p_y^2 + m^2}. \quad (5)$$

**3D Case:** In three spatial dimensions, the momentum is  $\mathbf{p} = (p_x, p_y, p_z)$ , so  $|\mathbf{p}|^2 = p_x^2 + p_y^2 + p_z^2$ . The energy eigenvalues are:

$$E_{\pm}(\mathbf{p}) = \pm \sqrt{p_x^2 + p_y^2 + p_z^2 + m^2}. \quad (6)$$

The general form presented in Equation (3) encapsulates these results for 1, 2, and 3 dimensions. These expressions provide the energy dispersion relations required for Problem 1.

## 1.1 Eigenstates, Berry Connection, and Curvature for the 2D Hamiltonian

We now focus on the 2D Hamiltonian, which can be written as  $H = \mathbf{h}(\mathbf{p}) \cdot \vec{\sigma}$ , where  $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$  are the Pauli matrices and  $\mathbf{h}(\mathbf{p}) = (p_x, p_y, m)$ , assuming  $v = 1$ . The magnitude of  $\mathbf{h}$  is  $h = |\mathbf{h}| = \sqrt{p_x^2 + p_y^2 + m^2}$ . The energy eigenvalues are  $E_{\pm} = \pm h$ .

The normalized eigenstate corresponding to the positive energy  $E_+ = h$  can be found using projection operators or by direct diagonalization. It is given by:

$$\psi_{E_+}(\mathbf{p}) = \frac{1}{\sqrt{2h(h+m)}} \begin{pmatrix} m+h \\ p_x + ip_y \end{pmatrix}. \quad (7)$$

(Note: The  $p_x - ip_y$  term in the original prompt for  $\psi_{E_+}$  corresponds to  $H = p_x\sigma_x - p_y\sigma_y + m\sigma_z$  or if  $\sigma_y$  definition is different. For  $H = p_x\sigma_x + p_y\sigma_y + m\sigma_z$ , the term is typically  $p_x + ip_y$ . We will proceed with the user's form  $p_x - ip_y$  for the Berry connection as provided, assuming it matches their Hamiltonian convention, though the standard eigenstate for  $H = p_x\sigma_x + p_y\sigma_y + m\sigma_z$  is  $\frac{1}{\sqrt{2h(h+m)}} \begin{pmatrix} h+m \\ p_x + ip_y \end{pmatrix}$ . If the Hamiltonian is  $H = p_x\sigma_x + p_y\sigma_y + m\sigma_z$  and the eigenstate is  $\psi_{E_+} = \frac{1}{\sqrt{2h(h+m)}} \begin{pmatrix} h+m \\ p_x - ip_y \end{pmatrix}$ , this implies a specific choice of Pauli matrices or a rotation. For the Berry connection components provided by the user, we use their eigenstate form.)

Let's use the eigenstate form that leads to the user's Berry connection:

$$\psi_{E_+}(\mathbf{p}) = \frac{1}{\sqrt{2h(h+m)}} \begin{pmatrix} m+h \\ p_x - ip_y \end{pmatrix}. \quad (8)$$

The eigenstate for  $E_- = -h$  is:

$$\psi_{E_-}(\mathbf{p}) = \frac{1}{\sqrt{2h(h-m)}} \begin{pmatrix} m-h \\ p_x - ip_y \end{pmatrix}. \quad (9)$$

The Berry connection  $\mathcal{A}(\mathbf{p}) = i\langle \psi_{E_+} | \nabla_{\mathbf{p}} \psi_{E_+} \rangle$  for the upper band  $\psi_{E_+}$  has components:

$$\mathcal{A}_x(\mathbf{p}) = \frac{-p_y}{2h(h+m)}, \quad \mathcal{A}_y(\mathbf{p}) = \frac{p_x}{2h(h+m)}. \quad (10)$$

The Berry curvature  $\Omega(\mathbf{p})$  (also denoted  $F_{xy}(\mathbf{p})$ ) is calculated as:

$$\Omega(\mathbf{p}) = \frac{\partial \mathcal{A}_y}{\partial p_x} - \frac{\partial \mathcal{A}_x}{\partial p_y}. \quad (11)$$

For this system, the Berry curvature is found to be:

$$\Omega(\mathbf{p}) = \frac{m}{2h^3} = \frac{m}{2(p_x^2 + p_y^2 + m^2)^{3/2}}. \quad (12)$$

Since  $E = h$  (with  $v = 1$ ), this can also be written as  $\Omega(\mathbf{p}) = \frac{m}{2E^3}$ .

The Chern number  $C$  is obtained by integrating the Berry curvature over the entire 2D momentum space (the Brillouin zone for a lattice system, or  $\mathbb{R}^2$  for a continuum model):

$$C = \int \frac{d^2 p}{2\pi} \Omega(\mathbf{p}). \quad (13)$$

Substituting the expression for  $\Omega(\mathbf{p})$  and converting to polar coordinates ( $p_x = p \cos \phi$ ,  $p_y = p \sin \phi$ ,  $d^2 p = p dp d\phi$ , where  $p = \sqrt{p_x^2 + p_y^2}$ ):

$$\begin{aligned} C &= \int_0^{2\pi} \frac{d\phi}{2\pi} \int_0^\infty p dp \frac{m}{2(p^2 + m^2)^{3/2}} \\ &= 1 \cdot \frac{m}{2} \int_0^\infty \frac{p}{(p^2 + m^2)^{3/2}} dp. \end{aligned} \quad (14)$$

To evaluate the integral, let  $u = p^2 + m^2$ , so  $du = 2p dp$ . When  $p = 0$ ,  $u = m^2$ . When  $p \rightarrow \infty$ ,  $u \rightarrow \infty$ .

$$\begin{aligned} C &= \frac{m}{2} \int_{m^2}^\infty \frac{1}{u^{3/2}} \frac{du}{2} = \frac{m}{4} \int_{m^2}^\infty u^{-3/2} du \\ &= \frac{m}{4} \left[ \frac{u^{-1/2}}{-1/2} \right]_{m^2}^\infty = \frac{m}{4} \left[ -2u^{-1/2} \right]_{m^2}^\infty \\ &= -\frac{m}{2} \left( \lim_{u \rightarrow \infty} \frac{1}{\sqrt{u}} - \frac{1}{\sqrt{m^2}} \right) \\ &= -\frac{m}{2} \left( 0 - \frac{1}{|m|} \right) = \frac{m}{2|m|} \\ &= \frac{1}{2} \text{sign}(m). \end{aligned} \quad (15)$$

Thus, the Chern number is  $C = +1/2$  if  $m > 0$  and  $C = -1/2$  if  $m < 0$ . If  $m = 0$ , the Chern number is ill-defined as the gap closes.

## 2 Problem 2: Edge States of 2D Dirac Hamiltonian

We consider a 2D Dirac Hamiltonian with a mass term  $m(y)$  that varies along the  $y$ -direction, creating an interface. Such a system can host chiral edge states, as described by Jackiw and Rebbi (1976) and found in topological insulators (e.g., Kane and Hazan review, Eq. 6).

The reference solution (Eq. 6 in Kane and Hazan) is of the form:

$$\psi_{q_x}(x, y) = \mathcal{N} e^{iq_x x} \exp\left(-\int_0^y \frac{m(y')}{\hbar v_F} dy'\right) \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (16)$$

with a linear energy dispersion relation:

$$E(q_x) = \hbar v_F q_x. \quad (17)$$

Here,  $q_x$  is the wavevector along the edge (x-direction),  $v_F$  is the Fermi velocity, and  $\mathcal{N}$  is a normalization constant. The mass  $m(y)$  changes sign across  $y = 0$  (e.g.,  $m(y) < 0$  for  $y < 0$  and  $m(y) > 0$  for  $y > 0$ ) for the state to be localized at the interface.

## 2.1 Derivation of the Edge State

To derive this solution, we consider a Hamiltonian:

$$H = v_F(\hat{p}_x \sigma_x - \hat{p}_y \sigma_y) + m(y) \sigma_z, \quad (18)$$

where  $\hat{p}_x = -i\hbar\partial_x$ ,  $\hat{p}_y = -i\hbar\partial_y$ , and  $\sigma_i$  are the standard Pauli matrices. We look for a solution of the form  $\psi(x, y) = e^{iq_x x} \Phi(y)$ . Substituting  $\hat{p}_x \rightarrow \hbar q_x$ :

$$H\psi = [v_F(\hbar q_x \sigma_x - (-i\hbar\partial_y) \sigma_y) + m(y) \sigma_z] e^{iq_x x} \Phi(y) = E e^{iq_x x} \Phi(y). \quad (19)$$

We propose an ansatz for the spinor part  $\Phi(y) = \phi(y)\chi$ , where  $\chi = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ . This spinor is an eigenstate of  $\sigma_x$  with eigenvalue +1:  $\sigma_x \chi = \chi$ . The eigenvalue equation becomes:

$$[v_F \hbar q_x \sigma_x + i\hbar v_F \sigma_y \partial_y + m(y) \sigma_z] \phi(y) \chi = E \phi(y) \chi. \quad (20)$$

We are looking for states with the dispersion  $E = \hbar v_F q_x$ . Substituting this and using  $\sigma_x \chi = \chi$ :

$$[\hbar v_F q_x I + i\hbar v_F \sigma_y \partial_y + m(y) \sigma_z] \phi(y) \chi = \hbar v_F q_x \phi(y) \chi. \quad (21)$$

This simplifies to:

$$(i\hbar v_F \sigma_y \partial_y + m(y) \sigma_z) \phi(y) \chi = 0. \quad (22)$$

Expanding  $\partial_y(\phi(y)\chi) = (\partial_y \phi)\chi$  (since  $\chi$  is  $y$ -independent):

$$i\hbar v_F (\partial_y \phi) \sigma_y \chi + m(y) \phi \sigma_z \chi = 0. \quad (23)$$

We calculate the action of Pauli matrices on  $\chi = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ :

$$\begin{aligned} \sigma_y \chi &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -i \\ i \end{pmatrix} = i \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \\ \sigma_z \chi &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \end{aligned}$$

Substituting these into the equation:

$$i\hbar v_F(\partial_y \phi) \begin{pmatrix} i \begin{pmatrix} -1 \\ 1 \end{pmatrix} \end{pmatrix} + m(y)\phi \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0. \quad (24)$$

$$-\hbar v_F(\partial_y \phi) \begin{pmatrix} -1 \\ 1 \end{pmatrix} + m(y)\phi \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0. \quad (25)$$

This can be rewritten as:

$$(\hbar v_F(\partial_y \phi) + m(y)\phi) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0. \quad (26)$$

For a non-trivial solution, we require the scalar part to be zero:

$$\hbar v_F \frac{d\phi}{dy} + m(y)\phi(y) = 0. \quad (27)$$

This is a first-order linear differential equation for  $\phi(y)$ :

$$\frac{d\phi}{dy} = -\frac{m(y)}{\hbar v_F}\phi(y). \quad (28)$$

The solution is:

$$\phi(y) = \mathcal{N} \exp \left( -\int_0^y \frac{m(y')}{\hbar v_F} dy' \right), \quad (29)$$

where  $\mathcal{N}$  is a normalization constant. Combining this with the ansatz, we recover the edge state wavefunction Eq. (6)

**Normalization and Localization:** For the state to be a localized edge state,  $\phi(y)$  must decay as  $y \rightarrow \pm\infty$ . This occurs if the integral  $\int_0^y \frac{m(y')}{\hbar v_F} dy'$  tends to  $+\infty$  as  $y \rightarrow \infty$  and to  $-\infty$  as  $y \rightarrow -\infty$  (or vice-versa, depending on the sign convention of  $m(y)$ ). Consider a mass domain wall where  $m(y) = m_0 \text{sgn}(y)$  with  $m_0 > 0$ . Then  $\int_0^y \frac{m(y')}{\hbar v_F} dy' = \frac{m_0}{\hbar v_F} |y|$ . The wavefunction component becomes  $\phi(y) = \mathcal{N} \exp \left( -\frac{m_0 |y|}{\hbar v_F} \right)$ . This state is localized at  $y = 0$  and is normalizable. The condition from the reference image ( $m(y) > 0$  for  $y > 0$  and  $m(y) < 0$  for  $y < 0$ ) ensures such localization.

**Fractionalization of fermion:** Fermion number fractionalization occurs when a topological defect, such as a mass domain wall in a Dirac system, hosts a localized zero-energy electronic state. This special state, positioned exactly between occupied negative-energy states (like the Dirac sea) and empty positive-energy states, results in the defect effectively binding a non-integer fermion number, typically  $\pm 1/2$ , compared to a system without such a defect. Consequently, the defect itself appears to carry a fractional charge or fermion number within the material's collective electronic ground state, even though fundamental fermions remain indivisible.

## 2.2 Dispersion Plot for Bulk and Edge States

The following plot illustrates the bulk energy bands for a 2D Dirac material and the linearly dispersing chiral edge state that appears at an interface where the mass term  $m$  changes sign. We set  $\hbar = 1$ ,  $v_F = 1$ , and a representative mass  $m = 0.5$  for the bulk gap. The edge state dispersion is  $E = q_x$ .

### 3 Summery

Berry curvature, Berry connection, Chern number: The Berry connection describes the geometric phase acquired by a quantum state as parameters are varied adiabatically; its curl is the Berry curvature, and the integral of this curvature over the Brillouin zone gives the Chern number, a topological invariant characterizing electronic bands.

Bulk-Boundary correspondence: This principle states that the non-trivial topological properties of a material's bulk (e.g., a non-zero Chern number) necessitate the existence of protected, gapless states at its boundaries or edges.

Edge states and why are they robust: Edge states are electronic states localized at the boundary of a material, often with unique transport properties; they are robust because their existence is guaranteed by the bulk topology, making them insensitive to local perturbations that do not close the bulk gap or break fundamental symmetries.

Explain how in Solid state systems the Dirac equation can appear and why it can have both negative and positive mass: In some solid-state systems (like graphene or topological insulators), the electron energy-momentum relationship near certain points (Dirac points) becomes linear, mimicking the massless Dirac equation; a gap can be opened, leading to an effective mass term which can be positive or negative depending on the band structure details (e.g., band inversion), distinguishing trivial from topological phases.